

TRANSFORMING FINANCIAL LEASING WITH MACHINE LEARNING: INTELLIGENT RISK ASSESSMENT AND PREDICTIVE ANALYTICS

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

The present individual component within an integrated machine learning framework targeting financial leasing addresses the computation of impairment amounts alongside the one-year Expected Credit Loss (ECL) in accordance with International Financial Reporting Standard 9 (IFRS 9). The impairment computation module adopts a data-driven methodology utilising 532,481 historical vehicle lease account records obtained from a Sri Lankan financial institution. The approach incorporates extensive feature construction, encompassing ratio-derived features, logarithmic transformations, polynomial terms, and interaction variables, all formulated to capture non-linear patterns inherent to lease repayment behaviour and credit exposure dynamics.

Six regression models were assessed: Random Forest, XGBoost, LightGBM, CatBoost, Gradient Boosting, and Ridge Regression with Polynomial Features. A Stacking Ensemble and an Optuna-optimised XGBoost variant were additionally examined. Experimental outcomes indicate that all ensemble-based models surpassed the 95% R^2 accuracy benchmark for Impairment prediction, with Random Forest attaining the highest score of 0.9912 (99.12%). For the one-year ECL prediction objective, R^2 values ranged between 0.6855 and 0.8487, reflecting the intrinsically higher complexity of estimating forward-looking losses. The Optuna-optimised XGBoost achieved a cross-validation R^2 of 0.9847 for Impairment following 50 trials. The top-performing model was serialised and exposed via a FastAPI RESTful endpoint incorporated into the broader system architecture.

Keywords: Impairment Estimation, Expected Credit Loss, IFRS 9, Ensemble Learning, XGBoost, Random Forest, LightGBM, CatBoost, Feature Engineering, Financial Leasing

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CV	Cross-Validation
ECL	Expected Credit Loss
EAD	Exposure at Default
FAISS	Facebook AI Similarity Search
FastAPI	Fast Application Programming Interface (Python Web Framework)
IFRS 9	International Financial Reporting Standard 9
LGD	Loss Given Default
LightGBM	Light Gradient Boosting Machine
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NPL	Non-Performing Loan / Lease
PD	Probability of Default
RAG	Retrieval-Augmented Generation
RMSE	Root Mean Square Error
R ²	Coefficient of Determination (R-Squared)
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
TPE	Tree-structured Parzen Estimator
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION

1.1 Background and Overview

Within the economies of developing nations such as Sri Lanka, the financial leasing sector occupies a central position by enabling the acquisition of income-generating assets including commercial vehicles, machinery, and capital equipment through structured payment arrangements. Unlike conventional lending products, financial leases bind a lessee to periodic rental payments to a lessor over an agreed duration, with the leased asset functioning as security. The financial viability of leasing companies is therefore intrinsically tied to the credit quality of their lease books and the precision with which impairment is measured.

Impairment denotes the shortfall between a financial asset's carrying value and its recoverable amount, arising when a borrower's capacity or willingness to repay deteriorates. Within financial leasing, impairment emerges when a lessee defaults on scheduled rental instalments, resulting in arrears accumulation that may ultimately necessitate account write-off. Beyond serving as a cornerstone of prudent risk management, accurate impairment measurement also constitutes a mandatory regulatory obligation under internationally recognised accounting standards.

A pivotal transformation in impairment accounting came with the issuance of International Financial Reporting Standard 9 (IFRS 9) by the International Accounting Standards Board (IASB), which fundamentally overhauled how financial institutions recognise credit losses. The previous framework under IAS 39 relied on an incurred loss paradigm, acknowledging impairment only after a triggering credit event had occurred. This retrospective methodology attracted widespread criticism for systematically underestimating credit losses in benign economic periods and amplifying financial fragility during crises, as witnessed during the 2007–2009 global financial turmoil [1]. IFRS 9 supplanted this approach with a forward-looking expected credit loss (ECL) model that obliges institutions to anticipate future losses drawing on historical evidence, current conditions, and plausible projections of future macroeconomic developments.

Within the IFRS 9 framework, ECL is determined as a product of three core risk parameters: the Probability of Default (PD), quantifying the chance of repayment failure; the Loss Given Default (LGD), measuring the irrecoverable fraction of exposure should default occur; and the Exposure at Default (EAD), capturing the outstanding contractual obligation at the point of default. The one-year ECL specifically measures credit losses expected from potential default occurrences within the next twelve months and applies to Stage 1 instruments under IFRS 9's three-stage categorisation structure [2].

Conventional approaches to impairment estimation including logistic regression and linear discriminant analysis have revealed considerable shortcomings in modelling the intricate, non-linear interdependencies present in large lease portfolios [3]. These approaches are constrained by linearity and normality assumptions, rendering them unable to adequately capture temporal patterns in repayment behaviour, heterogeneity among borrowers, and the multidimensional character of credit risk. Accordingly, a growing body of evidence indicates that machine learning methods can deliver substantially better predictive performance for impairment estimation in financial services [4].

This individual research effort is centred on building, implementing, and evaluating a machine learning-based impairment estimation module forming part of a comprehensive AI platform for financial leasing risk management at Sri Lanka Institute of Information Technology. The wider research initiative, pursued collectively by four researchers, encompasses four distinct sub-systems: default risk prediction, impairment and ECL estimation, branch performance analytics, and a retrieval-augmented generation (RAG)-powered conversational assistant. The impairment estimation module represents this author's principal individual contribution.

1.2 Research Problem

Leasing institutions operating in Sri Lanka contend with an enduring challenge in precisely quantifying and provisioning for impairment across their lease portfolios. The current landscape is characterised by several distinct deficiencies:

- Incurred loss frameworks under IAS 39 and simplified ECL approximations are inherently unable to reflect the forward-looking nature of credit deterioration, resulting in chronic underprovisioning during periods of portfolio growth.
- Computing impairment provisions manually is a resource-intensive process that is both slow and prone to human error, particularly when portfolios extend to hundreds of thousands of individual lease accounts.
- Conventional regression approaches applied to lease data fail to adequately capture non-linear interdependencies among predictor variables such as arrears duration, outstanding amounts, effective interest rates, and residual tenor and their impact on resulting impairment values.
- Without a robust feature construction process, the informational value embedded in raw transaction data such as the combined influence of arrears behaviour and interest rate exposure remains unexploited for predictive modelling purposes [5].
- Portfolio managers are without data-driven mechanisms for early detection of accounts facing significant impairment risk, constraining the institution's capacity to execute timely intervention strategies.

The central research question this study addresses is: In what ways can machine learning regression models, augmented by advanced feature engineering, be leveraged to accurately estimate impairment amounts and one-year ECL values for individual lease accounts within a Sri Lankan financial leasing environment, consistent with IFRS 9 provisions?

1.3 Research Objectives

The objectives established for the impairment estimation component of this research are as follows:

1. To construct a systematic feature engineering framework that converts raw vehicle leasing transaction data into informative predictor variables for use in impairment and ECL estimation.

2. To build and evaluate six regression models Random Forest, XGBoost, LightGBM, CatBoost, Gradient Boosting, and Ridge Regression with Polynomial Features for predicting both impairment amounts and one-year ECL.
3. To measure model performance through standard evaluation metrics comprising R^2 , RMSE, MAE, and MAPE, using a primary performance benchmark of $R^2 \geq 0.95$.
4. To apply hyperparameter optimisation to XGBoost via the Optuna framework and determine whether automated tuning produces measurable accuracy gains relative to the manually configured model.
5. To construct a Stacking Ensemble regressor that aggregates the outputs of multiple base models and assess whether this aggregation strategy enhances overall predictive quality.
6. To serialise and deploy the best-performing model as a production-ready artifact accessible through the project's RESTful API backend.

Objective	Description	Expected Outcome
1	Feature Engineering	Derived features capturing arrears, payment burden, interest exposure
2	Model Implementation	Six regression models trained on 80/20 train-test split
3	Performance Evaluation	$R^2 \geq 0.95$ for Impairment; comprehensive metric reporting
4	Hyperparameter Tuning	Optuna-optimised XGBoost with 50 trials
5	Stacking Ensemble	Ensemble combining RF, XGBoost, LightGBM, CatBoost, GB
6	Deployment	Serialised .pkl model, FastAPI endpoint

Table 1.1: Summary of Research Objectives and Expected Outcomes

1.4 Significance of the Study

This study makes several notable contributions spanning financial technology, credit risk management, and applied machine learning, with particular relevance to the Sri Lankan context.

From a regulatory standpoint, the creation of an IFRS 9-aligned ECL estimation tool constitutes a direct response to mandatory compliance obligations placed on financial institutions. The accuracy of ECL computations directly affects the reliability of provisions recorded in financial statements, with downstream consequences for reported profitability, capital adequacy, and investor trust [1].

From an operational risk perspective, account-level impairment prediction empowers portfolio managers to proactively flag at-risk accounts, direct remediation resources efficiently, and provision more accurately. This represents a substantive shift from reactive, portfolio-wide impairment review towards anticipatory, account-level monitoring [6].

From a scholarly standpoint, the study delivers empirical insights into the relative performance of contemporary ensemble learning techniques applied to vehicle leasing data a domain that has attracted comparatively scant academic attention relative to traditional banking loan datasets [3].

1.5 Scope and Delimitations

The scope of this investigation is delimited as follows. The data originates exclusively from a single Sri Lankan financial institution and relates specifically to vehicle leasing products; extrapolation of findings to other asset classes, geographical regions, or institutional contexts requires further empirical validation. The study is performed on a static, historical dataset and excludes real-time data ingestion or continuous model updating. Macroeconomic covariates, commonly incorporated in IFRS 9 implementations, are absent owing to data availability limitations. The LGD and EAD parameters of the ECL formula are not independently modelled; instead, impairment is predicted directly from observed historical outcomes.

2. LITERATURE REVIEW

2.1 IFRS 9 and Expected Credit Loss Frameworks

Academic literature has extensively examined the replacement of IAS 39's incurred loss framework with the forward-looking ECL model prescribed by IFRS 9. Breed et al. outline a methodology for computing IFRS 9 ECL estimates in a forward-looking manner, emphasising the indispensable role of macroeconomic and macroprudential data in generating unbiased loss projections [2]. Their research illustrates how embedding macroeconomic scenarios into ECL computations substantially alters both the magnitude and timing of recognised impairment losses, especially under conditions of economic stress. The authors conclude that institutions basing ECL solely on historical loss patterns without forward-looking adjustments produce systematically biased and non-compliant estimates.

Mlinaric et al. investigate the practical consequences of adopting the expected loss model within the Portuguese banking sector, documenting that IFRS 9 adoption drove a material increase in provisioning relative to IAS 39 requirements, with notable effects on Tier 1 capital ratios and profitability [7]. These findings underscore the financial significance of ECL model accuracy, given that even marginal prediction errors can translate into meaningful discrepancies in reported financial positions thus reinforcing the case for machine learning methods capable of more precisely modelling loss-driving relationships.

IFRS 9's three-stage classification system is foundational to understanding ECL measurement. Stage 1 instruments, where no material deterioration in credit quality has occurred since origination, require recognition of a twelve-month ECL. Stage 2 assets, having exhibited a significant credit quality decline but not yet defaulted, necessitate lifetime ECL recognition. Stage 3, comprising defaulted or credit-impaired assets, similarly demands lifetime ECL provisioning. Accurate staging and ECL computation hinge on the reliability of the underlying PD, LGD, and EAD models [2]. Vehicle leasing portfolios introduce layer-specific complexities into ECL modelling due to characteristics unique to the underlying assets. Patterns of asset depreciation, resale values of repossessed vehicles in secondary markets, and seasonal lending activity all affect the EAD and LGD parameters. Consequently, model development

for leasing contexts requires domain-specific tailoring rather than direct adaptation of methodologies originally designed for banking loan products.

2.2 Machine Learning in Credit Risk and Impairment Estimation

Machine learning applications in credit risk assessment have drawn considerable research interest over the past decade. Noriega et al. conducted a systematic review of machine learning approaches to credit risk prediction, encompassing 87 studies published between 2012 and 2022 [3]. Their findings show that ensemble-based techniques particularly Random Forest and gradient boosting variants consistently surpass classical statistical methods such as logistic regression and discriminant analysis across diverse datasets and evaluation measures. This superiority is attributed to ensemble methods' ability to model non-linear feature interactions and to handle high-dimensional, heterogeneous input data without imposing linearity or normality constraints.

Suhadolnik et al. supply empirical confirmation of machine learning's superiority over conventional credit scoring approaches, showing that XGBoost achieves notable accuracy improvements over logistic regression benchmarks [4]. Their analysis underscores the importance of careful feature selection and engineering in determining model performance, indicating that architectural choices alone are insufficient without thoughtful predictor construction.

Bitetto et al. investigate machine learning and deep learning approaches to credit risk modelling, finding that tree-based ensemble methods achieve results comparable to deeper neural network architectures while offering substantially better interpretability [8]. This transparency advantage is especially important in the IFRS 9 regulatory context, where auditors and supervisors may require explanations for individual model predictions.

Jadhav et al. explore data-driven approaches to loan default prediction, demonstrating that gradient-boosted models trained on financial transaction data attain high predictive accuracy when paired with domain-specific feature engineering [9]. Niu et al. evaluate both machine learning and deep learning for credit risk estimation, finding that XGBoost outperforms neural network architectures when dataset size is moderate and the feature space is constrained [10].

2.3 Ensemble Learning Approaches

Ensemble learning constitutes a class of machine learning methodologies that aggregate predictions from multiple constituent learners to yield a more accurate and robust combined prediction than any single model in isolation. The theoretical basis rests on the bias-variance decomposition of prediction error, which establishes that combining diverse models can concurrently reduce both components of generalisation error.

Random Forest builds an ensemble of decision trees, each trained on a bootstrap-sampled subset of the training data, with randomised feature selection at each split node to promote diversity among trees [11]. The aggregate prediction for regression tasks is computed as the mean across all constituent tree outputs. Random Forest has been widely validated in financial risk contexts, with Jadhav et al. demonstrating its superiority over logistic regression for credit risk classification under class imbalance conditions [12].

XGBoost implements a regularised gradient boosting procedure that optimises a differentiable objective function through second-order Taylor expansion [13]. Built-in L1 and L2 regularisation terms applied to leaf node weights serve to manage model complexity. Nkuah et al. demonstrate that XGBoost attains state-of-the-art performance for loan default prediction, with F1 scores exceeding 0.93 on held-out evaluation data [14].

LightGBM adopts a leaf-wise tree growth strategy in contrast to the depth-wise approach of standard gradient boosting, facilitating faster convergence and frequently superior accuracy on large-scale datasets [15]. Its use of histogram-based binning of continuous variables accelerates training and reduces memory requirements. CatBoost, a gradient boosting implementation developed by Yandex, natively handles categorical features without manual preprocessing, employing an ordered boosting scheme that mitigates overfitting tendencies common to standard gradient boosting approaches [16].

Stacking Ensemble uses a meta-learner that is trained on the out-of-fold predictions of a set of base learners, enabling it to learn how to optimally weight base model contributions across different regions of the feature space. Bhattacharya and

Bhattacharya demonstrate the merits of stacking for credit default forecasting, showing that stacked configurations consistently surpass individual base models on ROC-AUC and F1 criteria [17].

Malik et al. compare machine learning approaches to credit risk modelling across participation and conventional banks, finding that ensemble methods uniformly outperform logistic regression across both institution types, with XGBoost achieving the strongest overall results [18].

2.4 Feature Engineering for Financial Data

Feature engineering is broadly acknowledged as one of the most consequential determinants of model performance in financial applications. Berg et al. establish that incorporating behavioural indicators derived from open banking transaction records substantially enhances the predictive power of credit scoring models, with SHAP-based analysis revealing that payment recency and transaction frequency features contribute more explanatory power than conventional credit bureau variables [19].

Henriksen et al. examine explainable AI techniques applied to credit assessment, finding through SHAP value analysis that interaction effects among borrower age, outstanding balance, and payment history are among the strongest drivers of credit default likelihood [20]. This lends theoretical justification to the inclusion of interaction terms within the feature engineering framework developed in the present study.

Garcia et al. assess machine learning for credit delinquency prediction in a developing country context, identifying that payment-behaviour-derived features including arrears duration, inter-payment intervals, and partial payment ratios are among the most discriminating predictors [21]. Logarithmic transformation of skewed financial variables is a well-established practice, as such variables frequently exhibit log-normal distributional properties; logarithmic scaling compresses the dynamic range of extreme values and reduces their undue influence on model fitting [22].

Carrera and Benalcazar find that polynomial feature augmentation substantially improves regression-based models' ability to predict payment probability across

accounts at different arrears stages [23]. Dominguez-Hurtado et al. further validate the effectiveness of LightGBM for credit rating prediction tasks in financial settings [24].

2.5 Research Gaps

Analysis of the existing literature reveals several noteworthy gaps that the present study is positioned to address. The preponderance of machine learning research in credit risk has concentrated on binary default prediction rather than the regression-based estimation of impairment amounts. Direct prediction of impairment as a continuous monetary quantity as required under IFRS 9 provisioning rules has received limited scholarly coverage.

Moreover, most published studies draw upon banking loan data from developed market contexts, which may not adequately represent the distinct structural attributes of vehicle leasing portfolios in emerging markets such as Sri Lanka. Vehicle leasing encompasses unique risk dimensions including asset depreciation dynamics, secondary market conditions, and bespoke repayment structures that are absent from conventional loan datasets.

Additionally, the concurrent prediction of both impairment amounts and one-year ECL from a shared feature set as implemented here has not been previously documented in the literature. Similarly, the use of automated hyperparameter search via Optuna for XGBoost-based impairment estimation has not been reported in the leasing domain. Furthermore, the integration of impairment estimation functionality within a production-oriented microservices architecture has not been previously demonstrated in prior academic work [25].

3. METHODOLOGY

3.1 Research Design

This study employs a quantitative, experimental research design grounded in secondary data analysis. The research adheres to a structured machine learning pipeline covering data acquisition, preprocessing, feature engineering, model development, performance evaluation, and deployment readiness. The experimental framework entails a controlled comparison of multiple regression algorithms applied to the same dataset under uniform preprocessing and evaluation protocols, enabling objective assessment of their relative merits.

The research is structured according to the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which delineates phases of business comprehension, data exploration, data preparation, modelling, evaluation, and deployment. A supervised learning paradigm is adopted, wherein historical impairment amounts and one-year ECL values constitute the response variables. The experimental protocol employs an 80%/20% training-test partition, supplemented by five-fold cross-validation applied to the training subset.

3.2 Dataset Description

The dataset employed in this study consists of 532,481 vehicle leasing records sourced from a Sri Lankan financial institution, covering the period 2025 to 2026 and encompassing a varied cross-section of the institution's active and historical lease portfolio. The complete dataset from the broader research project includes 532,482 vehicle leasing records across 30 primary attributes. Figure 3.1 depicts the unprocessed dataset as viewed in Microsoft Excel prior to any preprocessing.

	A	B	C	D	E	F	G	H	I	J
1	Facility amount	Tenor	Due Date	Effec. Rate	Flat Rate	Net Rental	No of Rental in arrears	Age	Impairment	1 yr ECL
2	3,985,108	60	5	21.99	13.14	110,042	1.00	1.00	24,993	24,993
3	1,782,000	48	1	19.25	11.04	53,517	2.00	2.00	2,691	2,691
4	160,000	24	22	37.99	22.14	9,618	22.19	84.46	170,825	-
5	174,950	38	22	30.55	12.79	5,370	11.11	78.38	68,578	-
6	133,900	31	8	27.00	14.98	4,837	23.89	71.60	88,220	-
7	133,900	37	1	27.00	17.77	4,641	14.45	52.42	58,479	-
8	184,600	37	1	27.00	16.87	6,256	22.35	56.31	105,592	-
9	700,000	60	22	15.50	8.87	16,838	2.45	2.45	4,263	4,263
10	1,058,000	60	5	16.00	9.77	26,249	1.87	1.87	6,299	6,299
11	85,000	12	1	34.00	31.79	9,335	2.72	40.69	27,023	-
12	230,000	36	5	35.00	25.80	11,334	30.85	33.69	225,930	-
13	280,000	36	5	35.00	26.16	13,882	1.59	4.42	3,354	3,354
14	335,326	18	8	32.00	21.71	24,696	17.99	35.73	339,999	-
15	370,000	48	8	32.00	23.47	14,946	34.99	34.99	379,291	-
16	308,000	36	8	34.00	24.87	14,939	0.00	0.00	9	9
17	340,000	36	8	33.00	23.76	16,177	2.63	2.63	369	369
18	300,000	36	8	32.00	21.99	13,830	0.34	0.34	40	40
19	250,000	48	8	34.00	24.73	10,360	1.42	1.42	3,905	3,905
20	360,000	36	15	33.00	23.48	17,043	24.78	24.78	299,592	-
21	308,000	42	15	34.00	24.38	13,592	0.00	0.00	6	6
22	375,000	48	15	34.00	24.15	15,358	1.98	1.98	6,027	6,027
23	250,000	36	15	33.00	24.07	11,958	1.04	1.04	106	106
24	300,000	42	15	34.00	24.74	13,328	0.35	0.35	209	209
25	275,000	36	15	34.00	25.55	13,493	0.27	0.27	37	37
26	269,000	48	15	34.00	25.87	11,404	0.00	0.00	606	606
27	300,000	48	15	22.00	15.60	10,149	0.40	0.40	3,032	3,032
28	130,000	24	15	34.00	27.82	8,430	14.96	26.44	108,534	-
29	290,000	36	15	34.00	24.75	14,037	4.45	4.45	11,505	11,505
30	200,000	36	15	32.00	22.56	9,316	8.23	8.23	14,877	11,070

Figure 3.1: Raw Dataset Structure Excel View (data.xlsx)

Feature	Type	Description
Facility Amount	Numerical	Total leased asset value (LKR)
Tenor	Numerical	Lease duration in months
Due Date	Date	Scheduled lease maturity date
Effec. Rate	Numerical	Effective annual interest rate (%)
Flat Rate	Numerical	Flat annual interest rate (%)
Net Rental	Numerical	Monthly rental installment amount (LKR)
No of Rental in Arrears	Numerical	Count of missed rental payments
Age	Numerical	Age of lease account (months)
Impairment	Numerical (Target)	Observed impairment amount (LKR)
1 yr ECL	Numerical (Target)	One-year expected credit loss (LKR)

Table 3.1: Dataset Feature Overview

3.3 Data Preprocessing

The preprocessing pipeline consists of multiple sequential stages designed to safeguard data quality, address missing observations, and prepare the feature set for model training. Figure 3.2 shows the Python library imports used during implementation, and Figure 3.3 displays the data loading and preprocessing code.

```
] : import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split, cross_val_score, KFold
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.pipeline import Pipeline
from sklearn.base import clone
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor, StackingRegressor
from sklearn.linear_model import Ridge
from xgboost import XGBRegressor
from lightgbm import LGBMRegressor
from catboost import CatBoostRegressor
from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error
import matplotlib.pyplot as plt
import seaborn as sns
import optuna
import joblib
import warnings
warnings.filterwarnings('ignore')
optuna.logging.set_verbosity(optuna.logging.WARNING)
```

Figure 3.2: Python Library Imports for Model Implementation

```
: # Load data
print("\n[1/7] Loading data...")
file_path = 'data.xlsx' # Change to your file path
df = pd.read_excel(file_path)
print(f"✓ Data loaded: {df.shape[0]:,} rows, {df.shape[1]} columns")

# Data overview
print(f"\nColumn dtypes breakdown:\n{df.dtypes.value_counts().to_string()}")
print(f"Total missing values: {df.isnull().sum().sum()}")
missing = df.isnull().sum()
if missing.any():
    print(f"Columns with missing values:\n{missing[missing > 0].to_string()}")

# Data preprocessing
print("\n[2/7] Preprocessing data...")
if 'Due Date' in df.columns:
    df['Due Date'] = pd.to_datetime(df['Due Date'], errors='coerce')
    df['Days_to_Due'] = (df['Due Date'] - pd.Timestamp.now()).dt.days
    df['Months_to_Due'] = df['Days_to_Due'] / 30
    df['Years_to_Due'] = df['Days_to_Due'] / 365
    df = df.drop('Due Date', axis=1)

# Handle missing values
df = df.fillna(df.median(numeric_only=True))
print(f"✓ Missing values filled with column medians")
```

Figure 3.3: Data Loading and Preprocessing Pipeline Code

3.3.1 Missing Value Treatment

Missing values in numerical attributes were addressed through median imputation, replacing absent observations with the median value computed from training data for each respective feature. The median was preferred over the mean because of the positively skewed distributions characterising several financial variables. Missingness was minimal across the dataset: FlatRate at 0.16%, Equipment Type at 0.60%, and Last Receipt Paid Amount at 0.62%, with no feature exceeding 5% missing observations. The Due Date field was parsed into datetime format for temporal feature derivation (Days to Due, Months to Due, Years to Due), and the original column was subsequently dropped.

3.3.2 Data Splitting and Scaling

The 532,481 processed records were partitioned into training and evaluation subsets via stratified random sampling at an 80%/20% ratio, producing approximately 425,985 training observations and 106,496 test observations. All numerical attributes were then standardised using scikit-learn's StandardScaler, fitted exclusively on training data to prevent information leakage. The resulting scaling transformation was applied uniformly to both training and test subsets.

3.4 Feature Engineering

Feature engineering constitutes the most significant methodological contribution of this study. The framework was formulated to encode four distinct categories of information: repayment behaviour, financial exposure, temporal dynamics, and non-linear relationships. Figure 3.4 presents the complete feature engineering code.


```

# Feature Engineering
print("\n[3/7] Engineering features...")

# Create ratio and interaction features
df['Rate_Difference'] = df['Effec. Rate'] - df['Flat Rate']
df['Rental_to_Amount_Ratio'] = df['Net Rental'] / (df['Facility amount'] + 1)
df['Amount_per_Tenor'] = df['Facility amount'] / (df['Tenor'] + 1)
df['Rental_per_Tenor'] = df['Net Rental'] / (df['Tenor'] + 1)
df['Arrears_Rate'] = df['No of Rental in arrears'] / (df['Tenor'] + 1)
df['Total_Payment'] = df['Net Rental'] * df['Tenor']
df['Payment_Capacity'] = df['Facility amount'] / (df['Total_Payment'] + 1)
df['Risk_Score'] = df['No of Rental in arrears'] * df['Effec. Rate'] / 100
df['Age_Tenor_Interaction'] = df['Age'] * df['Tenor']
df['Amount_Rate_Interaction'] = df['Facility amount'] * df['Effec. Rate'] / 100
df['Arrears_Amount'] = df['No of Rental in arrears'] * df['Net Rental']

# Logarithmic features for skewed data
df['Log_Facility_Amount'] = np.log1p(df['Facility amount'])
df['Log_Net_Rental'] = np.log1p(df['Net Rental'])

# Squared features
df['Tenor_Squared'] = df['Tenor'] ** 2
df['Age_Squared'] = df['Age'] ** 2
df['Arrears_Squared'] = df['No of Rental in arrears'] ** 2

# Polynomial features for key variables
df['Rate_Squared'] = df['Effec. Rate'] ** 2
df['Rate_Cubed'] = df['Effec. Rate'] ** 3

print(f"✓ Features created: {df.shape[1]} total features")

```

Figure 3.4: Feature Engineering Code Implementation

3.4.1 Ratio and Interaction Features

Eleven ratio and interaction variables were derived to encode the financial relationships among the available raw attributes:

- Rate Difference: Effective Rate minus Flat Rate, reflecting the complexity of the interest pricing structure.
- Rental to Amount Ratio: Net Rental divided by (Facility Amount plus 1), expressing repayment burden relative to total credit exposure.
- Amount per Tenor: Facility Amount divided by (Tenor plus 1), approximating average monthly principal repayment.
- Rental per Tenor: Net Rental divided by (Tenor plus 1), representing average monthly installment per tenor unit.
- Arrears Rate: No. Arrears divided by (Tenor plus 1), expressing the fraction of total payments that have been missed.

- Total Payment: Net Rental multiplied by Tenor, yielding the aggregate contractual payment commitment.
- Payment Capacity: Facility Amount divided by (Total Payment plus 1), measuring the relationship between asset value and cumulative payment obligation.
- Risk Score: (No. Arrears multiplied by Effective Rate) divided by 100, forming a composite delinquency indicator by combining arrears count with financing cost.
- Age Tenor Interaction: Age multiplied by Tenor, capturing the interplay between account age and contracted duration.
- Amount Rate Interaction: (Facility Amount multiplied by Effective Rate) divided by 100, quantifying aggregate interest cost exposure.
- Arrears Amount: No. Arrears multiplied by Net Rental, representing the total monetary value of unpaid instalments.

3.4.2 Logarithmic Features

Two logarithmic transforms were computed to correct the right-skewed distributions of key financial variables, applying the formula $x_{log} = \log(1 + x)$ to accommodate zero-valued observations: Log Facility Amount and Log Net Rental.

3.4.3 Polynomial Features

Squared and cubed versions of critical features were derived to enable modelling of quadratic and cubic relationships: Tenor Squared ($Tenor^2$), Age Squared (Age^2), Arrears Squared ($No. Arrears^2$), Rate Squared ($Effective Rate^2$), and Rate Cubed ($Effective Rate^3$).

Feature Name	Formula / Derivation	Category
Rate Difference	Effec. Rate – Flat Rate	Ratio
Rental to_Amount_Ratio	Net Rental / (Facility Amount + 1)	Ratio
Amount_per_Tenor	Facility Amount / (Tenor + 1)	Ratio
Rental_per_Tenor	Net Rental / (Tenor + 1)	Ratio
Arrears Rate	No. Arrears / (Tenor + 1)	Ratio

Total Payment	$\text{Net Rental} \times \text{Tenor}$	Derived
Payment Capacity	$\text{Facility Amount} / (\text{Total Payment} + 1)$	Derived
Risk_Score	$(\text{No. Arrears} \times \text{Effec. Rate}) / 100$	Interaction
Age_Tenor_Interaction	$\text{Age} \times \text{Tenor}$	Interaction
Amount_Rate_Interaction	$(\text{Facility Amount} \times \text{Effec. Rate}) / 100$	Interaction
Arrears_Amount	$\text{No. Arrears} \times \text{Net Rental}$	Interaction
Log_Facility_Amount	$\log(1 + \text{Facility Amount})$	Logarithmic
Log_Net_Rental	$\log(1 + \text{Net Rental})$	Logarithmic
Tenor_Squared	Tenor^2	Polynomial
Age_Squared	Age^2	Polynomial
Arrears_Squared	No. Arrears^2	Polynomial
Rate_Squared	Effec. Rate^2	Polynomial
Rate_Cubed	Effec. Rate^3	Polynomial

Table 3.2: Engineered Features for Impairment and ECL Prediction

3.5 Model Development

Six regression models were built and assessed for both Impairment and one-year ECL prediction objectives. The model selection spans a broad spectrum of algorithmic paradigms, ranging from linear penalised regression to individual tree-based methods and sophisticated gradient boosting and stacking architectures.

3.5.1 Random Forest Regressor

The Random Forest Regressor was instantiated with 200 constituent decision trees ($n_estimators=200$), a maximum tree depth of 20 ($max_depth=20$), and scikit-learn default settings for all remaining parameters. Individual trees are trained on bootstrap-sampled subsets of the training data, with a random feature subset evaluated at each split node to ensure inter-tree diversity. The ensemble's final prediction is the mean of all 200 individual tree outputs [11].

3.5.2 XGBoost Regressor

XGBoost was configured using 200 estimators, a maximum depth of 10, a learning rate of 0.1, and default values for remaining hyperparameters. Its regularised gradient

boosting algorithm minimises a differentiable loss function via second-order Taylor expansion, incorporating built-in L1 and L2 regularisation on leaf weights to govern model complexity [13].

3.5.3 LightGBM Regressor

LightGBM was configured with 200 estimators, a maximum depth of 15, and up to 50 leaves per tree. Its leaf-wise tree growth mechanism permits construction of deeper, more expressive trees than conventional depth-wise gradient boosting approaches. Histogram-based binning of continuous features supports efficient training on large datasets [15].

3.5.4 CatBoost Regressor

CatBoost was initialised with 200 iterations, a maximum depth of 10, a learning rate of 0.1, and verbosity disabled. Its ordered boosting procedure addresses the prediction shift artefact that may affect standard gradient boosting implementations [16].

3.5.5 Gradient Boosting Regressor

The scikit-learn Gradient Boosting Regressor was configured with 200 estimators, a maximum depth of 10, and a learning rate of 0.1. This model serves as a reference gradient boosting implementation against which more advanced variants are benchmarked.

3.5.6 Ridge Regression with Polynomial Features

Ridge Regression with Polynomial Features serves as the linear reference model. The pipeline applies a Polynomial Features transformer (degree=2) followed by a Ridge Regressor (alpha=1.0). The Ridge penalty manages coefficient magnitudes to prevent overfitting within the resulting high-dimensional polynomial feature space.

3.5.7 Stacking Ensemble

A Stacking Ensemble was assembled using scikit-learn's StackingRegressor, designating Random Forest, XGBoost, LightGBM, CatBoost, and Gradient Boosting as base learners and a Ridge Regressor as the meta-learner. Out-of-fold cross-validation across five folds generates the meta-learner's training inputs, mitigating overfitting at the meta-learning stage [17].

Model	Key Hyperparameters	Algorithm Type
Random Forest	n_estimators=200, max_depth=20	Bagging Ensemble
XGBoost	n_estimators=200, max_depth=10, lr=0.1	Boosting Ensemble
LightGBM	n_estimators=200, max_depth=15, num_leaves=50	Boosting Ensemble
CatBoost	iterations=200, max_depth=10, lr=0.1	Boosting Ensemble
Gradient Boosting	n_estimators=200, max_depth=10, lr=0.1	Boosting Ensemble
Ridge (Polynomial)	degree=2, alpha=1.0	Linear Regularised
Stacking Ensemble	Base: RF, XGB, LGBM, CB, GB; Meta: Ridge	Stacking Ensemble

Table 3.3: Model Configurations for Regression Algorithms

3.6 Model Evaluation Metrics

Model performance was evaluated using four standard regression metrics calculated on the held-out test partition:

- Coefficient of Determination (R^2): Measures the proportion of variance in the target variable explained by the model. An R^2 of at least 0.95 was set as the primary performance criterion.
- Root Mean Square Error (RMSE): The square root of mean squared prediction error, expressed in LKR. This metric imposes greater penalties on large prediction errors through the squaring operation.
- Mean Absolute Error (MAE): The average absolute deviation between predicted and actual values in LKR, assigning equal weight to all errors irrespective of magnitude.
- Mean Absolute Percentage Error (MAPE): Expresses prediction error as a proportion of actual values. Reported for completeness but interpreted with caution given the presence of near-zero actual values in the distribution.

3.7 Hyperparameter Optimization

To determine whether automated tuning could surpass the manually configured XGBoost model, the Optuna framework was applied for a systematic search over the

XGBoost hyperparameter space. Optuna operates via Tree-structured Parzen Estimator (TPE) optimisation, a Bayesian method that constructs a probabilistic model of the objective surface and selects candidate configurations intelligently for evaluation.

The optimisation study encompassed 50 trials, with each trial assessing a candidate configuration using 5-fold cross-validation on the training set. The optimisation objective was the mean cross-validation R^2 score for Impairment prediction.

Hyperparameter	Search Space	Type
n_estimators	[100, 1000]	Integer
max_depth	[3, 10]	Integer
learning_rate	[0.01, 0.3]	Log-uniform Float
subsample	[0.5, 1.0]	Float
reg_lambda	[0.1, 10.0]	Log-uniform Float
min_child_weight	[1, 10]	Integer

Table 3.4: Optuna Hyperparameter Search Space for XGBoost

3.8 Deployment Architecture

The highest-performing Impairment prediction model was serialised via Python's joblib library and stored as a .pkl file. This stored artifact is loadable at inference time without necessitating model retraining. The API endpoint receives JSON-formatted inputs containing raw lease account feature values, applies the preprocessing and feature engineering pipeline, and returns predicted impairment amounts and one-year ECL in JSON format. Input validation is managed through Pydantic models within the FastAPI framework.

4. RESULTS AND DISCUSSION

4.1 Experimental Setup and Implementation Screenshots

This section documents the actual training output captured during experimental execution, serving as evidence of the model training processes and outcomes as run within the Python environment.

4.1.1 Random Forest Training Results

Figure 4.1 presents the training output for the Random Forest model for both Impairment and one-year ECL prediction tasks. The model achieved an R^2 of 0.9912 (99.12%) for Impairment, exceeding the 95% target, and an R^2 of 0.8448 (84.48%) for ECL, which fell below the target threshold. The five-fold cross-validation R^2 of 0.9896 ± 0.0013 for Impairment confirms stable generalization performance.

```
[5/7] Training models (this may take a few minutes)...
```

```
=====
Training: Random Forest
=====
```

```
--- IMPAIRMENT PREDICTION ---
```

```
Random Forest - Impairment:
R2 Score:      0.991195 (99.12%) ✓ TARGET ACHIEVED
Adjusted R2:    0.991192
RMSE:          20824.5320
MAE:           5431.9886
MAPE:          136545573446343.75%
CV R2 (5-fold): 0.9896 ± 0.0013
Model saved: random_forest_impairment.pkl
```

```
--- 1 YR ECL PREDICTION ---
```

```
Random Forest - 1 yr ECL:
R2 Score:      0.844826 (84.48%) ⚠ Below target
Adjusted R2:    0.844785
RMSE:          7695.6783
MAE:           3118.6664
MAPE:          403172555191873.88%
CV R2 (5-fold): 0.8411 ± 0.0023
Model saved: random_forest_ecl.pkl
```

Figure 4.1: Random Forest Impairment and ECL Training Results

4.1.2 XGBoost Training Results

Figure 4.2 presents the training output for the XGBoost model. XGBoost achieved an R^2 of 0.9906 (99.06%) for Impairment and 0.8355 (83.55%) for ECL. The five-fold cross-validation R^2 of 0.9860 ± 0.0047 for Impairment confirms reliable generalization.

```
=====
Training: XGBoost
=====

--- IMPAIRMENT PREDICTION ---

XGBoost - Impairment:
R2 Score:      0.990555 (99.06%) ✓ TARGET ACHIEVED
Adjusted R2:    0.990552
RMSE:          21568.2347
MAE:           6670.8745
MAPE:          190240281670431.09%
CV R2 (5-fold): 0.9860 ± 0.0047
Model saved: xgboost_impairment.pkl

--- 1 YR ECL PREDICTION ---

XGBoost - 1 yr ECL:
R2 Score:      0.835522 (83.55%) ⚠ Below target
Adjusted R2:    0.835479
RMSE:          7923.0348
MAE:           3511.5931
MAPE:          456998205928091.62%
CV R2 (5-fold): 0.8328 ± 0.0019
Model saved: xgboost_ecl.pkl
```

Figure 4.2: XGBoost Impairment and ECL Training Results

4.1.3 CatBoost Training Results

Figure 4.3 presents the CatBoost training output. CatBoost achieved an R^2 of 0.9905 (99.05%) for Impairment and 0.8214 (82.14%) for ECL. The five-fold cross-validation R^2 of 0.9834 ± 0.0047 for Impairment indicates consistent performance across folds.


```

=====
Training: CatBoost
=====

--- IMPAIRMENT PREDICTION ---

CatBoost - Impairment:
  R2 Score:      0.990504 (99.05%) ✓ TARGET ACHIEVED
  Adjusted R2:    0.990502
  RMSE:           21625.8542
  MAE:            7080.6134
  MAPE:           202551102947998.06%
  CV R2 (5-fold): 0.9834 ± 0.0047
  Model saved: catboost_impairment.pkl

--- 1 YR ECL PREDICTION ---

CatBoost - 1 yr ECL:
  R2 Score:      0.821444 (82.14%) ⚠ Below target
  Adjusted R2:    0.821397
  RMSE:           8255.1491
  MAE:            3722.0354
  MAPE:           482743585609673.81%
  CV R2 (5-fold): 0.8186 ± 0.0029
  Model saved: catboost_ecl.pkl

```

Figure 4.3: CatBoost Impairment and ECL Training Results

4.1.4 Gradient Boosting Training Results

Figure 4.4 presents the Gradient Boosting training output. Gradient Boosting achieved an R^2 of 0.9904 (99.04%) for Impairment and 0.8274 (82.74%) for ECL. The five-fold cross-validation R^2 of 0.9876 ± 0.0025 for Impairment indicates stable generalization.

```

=====
Training: Gradient Boosting
=====

--- IMPAIRMENT PREDICTION ---

Gradient Boosting - Impairment:
  R2 Score:      0.990398 (99.04%) ✓ TARGET ACHIEVED
  Adjusted R2:    0.990396
  RMSE:           21745.8192
  MAE:            7073.0521
  MAPE:           189899941775528.28%
  CV R2 (5-fold): 0.9876 ± 0.0025
  Model saved: gradient_boosting_impairment.pkl

--- 1 YR ECL PREDICTION ---

Gradient Boosting - 1 yr ECL:
  R2 Score:      0.827447 (82.74%) ⚠ Below target
  Adjusted R2:    0.827401
  RMSE:           8115.2037
  MAE:            3674.6660
  MAPE:           474170111850199.81%
  CV R2 (5-fold): 0.8241 ± 0.0019
  Model saved: gradient_boosting_ecl.pkl

```

Figure 4.4: Gradient Boosting Impairment and ECL Training Results

4.1.5 Optuna Hyperparameter Tuning Results

Figure 4.5 presents the Optuna tuning results for XGBoost. After 50 trials, the best trial (Trial 41) achieved a cross-validation R^2 of 0.9847 with the optimal configuration: $n_estimators=657$, $max_depth=5$, $learning_rate=0.0842$, $subsample=0.649$, $reg_lambda=4.514$, $min_child_weight=3$. The final test set R^2 was 0.9905 (99.05%) for Impairment.

```

print(" Model saved: xgboost_tuned_impairment.pkl")

results.append({
    'Model': 'XGBoost (Optuna Tuned)', 'Target': 'Impairment',
    'R² Score': r2_tuned, 'CV R² Mean': study.best_value, 'CV R² Std': 0.0,
    'Accuracy %': r2_tuned * 100, 'RMSE': rmse_tuned, 'MAE': mae_tuned, 'MAPE %': mape_tuned,
    'Status': '✓ Achieved' if r2_tuned * 100 >= 95 else '⚠ Below 95%'
})

[5c/7] Optuna hyperparameter tuning for XGBoost (Impairment)...
Running 50 trials – this may take a few minutes...
Best trial: 41. Best value: 0.984662: 100%|██████████| 50/50 [13:18<00:00, 15.97s/it]
✓ Best Optuna XGBoost CV R²: 0.984662
Best params: {'n_estimators': 657, 'max_depth': 5, 'learning_rate': 0.08421219718304562, 'subsample':
649298437, 'reg_lambda': 4.51425567267747, 'min_child_weight': 3}

XGBoost (Optuna Tuned) - Impairment:
R² Score: 0.990505 (99.05%) ✓ TARGET ACHIEVED
Adjusted R²: 0.990502
RMSE: 21625.0546
MAE: 6845.2514
MAPE: 198077011628121.59%
Model saved: xgboost_tuned_impairment.pkl

```

Figure 4.5: Optuna Hyperparameter Tuning XGBoost Best Trial Results

4.1.6 Complete Results Summary

Figure 4.6 presents the complete results summary generated by the model training pipeline, encompassing all models evaluated for both Impairment and one-year ECL prediction targets, along with their respective R² scores, CV R² means, accuracy percentages, RMSE, MAE, MAPE values, and achievement status relative to the 95% target.

```

[6/7] Generating results summary...

=====
COMPLETE RESULTS SUMMARY
=====

Model      Target  R² Score  CV R² Mean  Accuracy %  RMSE  MAE  MAPE %  Status
Stacking Ensemble  1 yr ECL  0.848742  NaN  84.874213  7597.956275  3185.447053  4.295452e+14  ⚠ Below 95%
Random Forest      1 yr ECL  0.844826  0.841096  84.482627  7695.678305  3118.666383  4.031726e+14  ⚠ Below 95%
LightGBM           1 yr ECL  0.840230  0.836659  84.022971  7808.827100  3471.040355  4.568372e+14  ⚠ Below 95%
XGBoost            1 yr ECL  0.835522  0.832789  83.552211  7923.034782  3511.593062  4.569982e+14  ⚠ Below 95%
Gradient Boosting  1 yr ECL  0.827447  0.824091  82.744671  8115.203677  3674.666002  4.741701e+14  ⚠ Below 95%
CatBoost           1 yr ECL  0.821444  0.818591  82.144408  8255.149111  3722.035370  4.827436e+14  ⚠ Below 95%
Ridge (Polynomial) 1 yr ECL  0.685495  0.682333  68.549540  10955.981720  5970.820225  7.048625e+14  ⚠ Below 95%
Random Forest Impairment 0.991195  0.989638  99.119477  20824.531962  5431.988603  1.365456e+14  ✓ Achieved
LightGBM Impairment 0.990924  0.988053  99.092362  21142.740447  6378.075607  1.740985e+14  ✓ Achieved
XGBoost Impairment 0.990555  0.986028  99.055463  21568.234717  6670.874462  1.902403e+14  ✓ Achieved
XGBoost (Optuna Tuned) Impairment 0.990505  0.984662  99.050479  21625.054600  6845.251436  1.980770e+14  ✓ Achieved
CatBoost Impairment 0.990504  0.983379  99.050409  21625.854222  7080.613398  2.025511e+14  ✓ Achieved
Gradient Boosting Impairment 0.990398  0.987627  99.039845  21745.819198  7073.052094  1.898999e+14  ✓ Achieved
Stacking Ensemble Impairment 0.990055  NaN  99.005525  22131.044481  5590.570410  1.545040e+14  ✓ Achieved
Ridge (Polynomial) Impairment 0.984941  0.980618  98.494137  27233.143366  10881.263523  1.640999e+14  ✓ Achieved

✓ Results saved: advanced_model_results.csv

```

Figure 4.6: Complete Results Summary All Models, Both Targets

4.2 Impairment Prediction Results

Every model configuration assessed for Impairment prediction surpassed the 95% R² threshold, with scores spanning from 0.9849 for Ridge Regression with Polynomial

Features to 0.9912 for Random Forest. Table 4.1 presents the comprehensive performance results.

Model	R ² Score	CV R ² Mean	Accuracy %	RMSE (LKR)	MAE (LKR)	Status
Random Forest	0.9912	0.9896	99.12%	20,824.53	5,431.99	✓ Achieved
LightGBM	0.9909	0.9881	99.09%	21,142.74	6,378.08	✓ Achieved
XGBoost	0.9906	0.9860	99.06%	21,568.23	6,670.87	✓ Achieved
XGBoost (Optuna)	0.9905	0.9847	99.05%	21,625.05	6,845.25	✓ Achieved
CatBoost	0.9905	0.9834	99.05%	21,625.85	7,080.61	✓ Achieved
Gradient Boosting	0.9904	0.9876	99.04%	21,745.82	7,073.05	✓ Achieved
Stacking Ensemble	0.9901	N/A	99.01%	22,131.04	5,590.57	✓ Achieved
Ridge (Polynomial)	0.9849	0.9806	98.49%	27,233.14	10,881.26	✓ Achieved

Table 4.1: Impairment Prediction Performance All Models

Random Forest ranked first for Impairment prediction with an R² of 0.9912 and a MAE of LKR 5,431.99. Its five-fold cross-validation R² of 0.9896 closely aligns with the test set R², confirming strong generalisation to unseen data. The low cross-validation variability (standard deviation of 0.0013) signals consistent performance across fold partitions [11].

Gradient boosting variants LightGBM, XGBoost, CatBoost, and Gradient Boosting produced R² values within the tight band of 0.9904 to 0.9909, attesting to the general effectiveness of gradient boosting architectures for tabular financial datasets [13]. LightGBM claimed the second-highest R² at 0.9909, validating the advantages of leaf-wise tree growth for this application [15].

The Stacking Ensemble recorded an R^2 of 0.9901, marginally below Random Forest and individual gradient boosting models. This outcome suggests the Ridge meta-learner may be suboptimally suited to synthesising base model predictions in this context, potentially benefiting from an alternative meta-learning strategy in subsequent work [17].

Ridge Regression with Polynomial Features, while attaining a commendable R^2 of 0.9849, reveals a notable performance disparity relative to non-linear models. Its considerably elevated RMSE (LKR 27,233.14) and MAE (LKR 10,881.26) compared to ensemble counterparts affirm the value of non-linear modelling capacity for impairment estimation tasks.

4.3 One-Year ECL Prediction Results

Table 4.2 presents the performance results for all models evaluated on the one-year ECL prediction task. The results reveal a markedly different pattern from the Impairment prediction task, with all models failing to achieve the 95% accuracy target ($R^2 \geq 0.95$) for ECL prediction.

Model	R^2 Score	CV R^2 Mean	Accuracy %	RMSE (LKR)	MAE (LKR)	Status
Stacking Ensemble	0.8487	N/A	84.87%	7,597.96	3,185.45	⚠ Below 95%
Random Forest	0.8448	0.8411	84.48%	7,695.68	3,118.67	⚠ Below 95%
LightGBM	0.8402	0.8367	84.02%	7,808.83	3,471.04	⚠ Below 95%
XGBoost	0.8355	0.8328	83.55%	7,923.03	3,511.59	⚠ Below 95%
Gradient Boosting	0.8274	0.8241	82.74%	8,115.20	3,674.67	⚠ Below 95%

CatBoost	0.8214	0.8186	82.14%	8,255.15	3,722.04	⚠ Below 95%
Ridge (Polynomial)	0.6855	0.6823	68.55%	10,955.98	5,970.82	⚠ Below 95%

Table 4.2: One-Year ECL Prediction Performance All Models

ECL prediction yielded R^2 values markedly below those observed for Impairment prediction (0.6855 to 0.8487 versus 0.9849 to 0.9912), indicating that the one-year ECL is intrinsically more difficult to estimate from the available feature set. This outcome aligns with the theoretical understanding of ECL as a forward-looking quantity that incorporates expectations regarding future economic circumstances and borrower conduct neither of which can be directly observed in historical transaction records [2].

The Stacking Ensemble attained the highest ECL prediction R^2 of 0.8487, in contrast to its comparatively lower performance on Impairment prediction. This reversal implies that ensemble aggregation delivers particular benefits for the more complex ECL estimation task, where individual models may complement each other by capturing different aspects of the relationship between input features and prospective credit losses [17].

Ridge Regression with Polynomial Features produced an R^2 of only 0.6855 for ECL prediction, further confirming that linear modelling approaches are inadequate for this task, where the stochastic and forward-looking nature of credit loss estimates demands greater model expressiveness.

4.4 Comparative Analysis

Table 4.3 presents a consolidated comparative summary of model performances across both prediction tasks.

Model	Impairment R^2	Impairment Acc%	ECL R^2	ECL Acc%	95% Target Met
Random Forest	0.9912	99.12%	0.8448	84.48%	Impairment only
LightGBM	0.9909	99.09%	0.8402	84.02%	Impairment only

XGBoost	0.9906	99.06%	0.8355	83.55%	Impairment only
XGBoost (Optuna)	0.9905	99.05%	N/A	N/A	Impairment only
CatBoost	0.9905	99.05%	0.8214	82.14%	Impairment only
Gradient Boosting	0.9904	99.04%	0.8274	82.74%	Impairment only
Stacking Ensemble	0.9901	99.01%	0.8487	84.87%	Impairment only
Ridge (Polynomial)	0.9849	98.49%	0.6855	68.55%	Impairment only

Table 4.3: Comparative Summary of All Model Performances

Across both prediction tasks, all models attained the 95% accuracy threshold for Impairment but fell short for ECL. The uniformity of this pattern across algorithmically diverse approaches implies that the difficulty of ECL prediction stems from the nature of the target variable rather than deficiencies in any particular model architecture.

The ranking of models diverges between the two tasks: Random Forest leads for Impairment while the Stacking Ensemble leads for ECL. This suggests that model selection should be performed independently for each target. The performance gap between tree-based ensembles and Ridge Regression is considerably wider for ECL (R^2 difference of approximately 0.16) than for Impairment (approximately 0.06), highlighting that non-linear modelling capacity matters more for the inherently more complex ECL objective.

4.4.1 XGBoost Optuna Tuned Results

Table 4.4 presents the detailed performance results for the Optuna-tuned XGBoost model for Impairment prediction.

Metric	Value
R ² Score	0.990505 (99.05%)
Adjusted R ²	0.990502
RMSE	LKR 21,625.05
MAE	LKR 6,845.25
Best CV R ² (Optuna)	0.984662
Best Trial	41 of 50

Optimal n_estimators	657
Optimal max_depth	5
Optimal learning_rate	0.0842
Optimal subsample	0.649
Optimal reg_lambda	4.514
Optimal min_child_weight	3

Table 4.4: XGBoost (Optuna Tuned) Impairment Prediction Results

The Optuna-optimised XGBoost achieved an R^2 of 0.9905 for Impairment prediction, marginally below the manually specified XGBoost ($R^2 = 0.9906$). This outcome where automated tuning fails to improve upon the manual baseline may reflect the fact that Optuna's cross-validation R^2 on training data does not perfectly align with the test set metric. The 50-trial search budget may also have been insufficient to thoroughly explore the six-dimensional hyperparameter space.

4.5 Discussion

4.5.1 Impairment Prediction as a Tractable Problem

The attainment of the 95% accuracy benchmark by every evaluated model for Impairment prediction confirms that machine learning-based impairment estimation constitutes a tractable problem, provided that appropriate feature engineering and sufficiently large training datasets are available [4]. The strong R^2 values achieved even by the linear Ridge Regression model (0.9849) indicate that a substantial share of impairment variance can be explained by linear relationships among the engineered predictors.

The feature engineering framework particularly the Risk Score interaction term (No. Arrears $\times$ Rate / 100), Arrears Amount, and the logarithmic transformations of facility amount and net rental appears to have been highly effective in encoding the primary impairment drivers. The consistently strong performance observed across all model types regardless of architecture suggests these engineered features are robustly predictive.

4.5.2 ECL Prediction as a More Complex Task

The failure of all models to meet the 95% R^2 target for ECL prediction reflects the inherent challenges of forward-looking credit loss estimation. One-year ECL captures anticipated borrower behaviour, economic trajectories, and asset value movements over a twelve-month horizon none of which are directly observable from historical lease transaction data [2]. The clustering of ECL prediction R^2 values around 0.82–0.85 for ensemble models suggests a performance ceiling attributable to the current feature set. Incorporating macroeconomic variables, forward-looking economic projections, and borrower-level credit data could potentially elevate ECL prediction accuracy.

4.5.3 Practical Implications

Based on experimental outcomes, Random Forest is the recommended primary model for Impairment prediction, given its highest R^2 score, lowest RMSE among interpretable model classes, and robust cross-validation results. For ECL prediction, the Stacking Ensemble yields the highest R^2 and may be preferred for deployment. In production, the model yields a MAE of LKR 5,431.99 for Impairment prediction, representing an error below 6% for accounts with impairment values of LKR 100,000 or above an operationally acceptable margin for provisioning purposes.

The notably elevated MAPE values for both targets arise from the influence of very small actual values in the denominator. For accounts with minimal impairment or ECL balances, even modest absolute errors produce disproportionately large percentage deviations. Consequently, MAPE should be interpreted cautiously in this application; MAE and RMSE are more informative performance indicators for the purpose at hand.

4.5.4 Comparison with Published Literature

The R^2 values achieved in this study (0.9849–0.9912 for Impairment) are among the highest reported for regression-based credit loss estimation in the literature. Jadhav et al. report F1 scores of 0.87–0.93 for gradient boosting on loan default prediction [9], while Bhattacharya and Bhattacharya report ROC-AUC values of 0.95–0.98 for stacking ensembles applied to credit default prediction [17]; both correspond qualitatively to comparable accuracy levels. Breed et al. similarly note that ECL estimation introduces additional uncertainty relative to point-in-time impairment

assessment when forward-looking macroeconomic information is required [2], consistent with the performance differential observed in this study.

5. CONCLUSION

5.1 Summary of Findings

This dissertation has documented the design, implementation, and empirical evaluation of a machine learning-based impairment estimation component for financial leasing risk management, representing the individual research contribution of Lasindu Semasingha to the broader integrated AI platform developed at Sri Lanka Institute of Information Technology.

The empirical results confirm that machine learning regression models, applied to a systematically engineered predictor set derived from historical vehicle leasing data, are capable of producing highly accurate account-level impairment predictions. All eight model configurations achieved R^2 values exceeding 0.98 for Impairment prediction, with the top-performing Random Forest reaching an R^2 of 0.9912 (99.12%) well above the 95% research target.

For one-year ECL prediction, no model attained the 95% target, with R^2 values ranging from 0.6855 (Ridge Regression with Polynomial Features) to 0.8487 (Stacking Ensemble). This finding reflects the inherently greater difficulty of forward-looking loss estimation relative to contemporaneous impairment prediction, highlighting the need for supplementary data sources such as macroeconomic variables to further improve ECL prediction accuracy [2].

The 18-feature engineering framework, spanning ratio, interaction, logarithmic, and polynomial feature categories, is identified as a key performance driver. The Risk Score interaction term, Arrears Amount, and the logarithmic transforms of facility amount and net rental proved particularly informative predictors.

Automated hyperparameter search using Optuna across 50 trials did not yield improvements over the manually configured XGBoost baseline for Impairment prediction (R^2 of 0.9905 versus 0.9906), suggesting the default configuration is near-optimal for this dataset and that further improvements may require an expanded search space or a larger trial budget.

5.2 Contribution

This research makes the following contributions across financial technology, credit risk management, and applied machine learning:

Feature Engineering Framework: A systematic, domain-tailored feature engineering methodology is presented for vehicle leasing impairment estimation, comprising 18 derived features spanning four categories. This framework serves as a domain-specific methodological contribution to financial leasing analytics and may act as a reference point for future investigations in comparable contexts.

Comparative Algorithm Evaluation: A rigorously controlled multi-model comparison covering six regression algorithms plus Optuna-tuned and stacking extensions applied to a real-world Sri Lankan vehicle leasing dataset of 532,481 records is provided. The empirical evidence enriches the limited scholarly literature on machine learning applied specifically to vehicle leasing portfolios.

IFRS 9 Implementation Insights: A practical demonstration of IFRS 9-compliant impairment and ECL estimation via machine learning is delivered, offering actionable implementation guidance for financial institutions in Sri Lanka and analogous developing markets [2][7].

Deployment Architecture: The embedding of the trained model within a RESTful API backend establishes the operational viability of machine learning-driven impairment estimation within enterprise-grade system architectures, narrowing the divide between research prototypes and production deployment.

5.3 Limitations

Several constraints of this study warrant acknowledgement. The dataset derives exclusively from one Sri Lankan financial institution and covers only vehicle leasing products; the degree to which the models generalise to alternative institutions, markets, or asset types remains untested. The present approach also excludes macroeconomic indicators commonly integrated into IFRS 9 ECL frameworks owing to data unavailability [2].

The models are trained on a static historical dataset and lack mechanisms for dynamic updating as new observations accumulate. Operational deployment would require

periodic retraining to accommodate shifts in portfolio composition, macroeconomic conditions, and evolving borrower behaviour patterns. Furthermore, the implementation predicts impairment as an aggregate outcome rather than modelling PD, LGD, and EAD as distinct ECL components, which constrains interpretability in terms of individual risk parameter contributions.

5.4 Future Work

Several avenues for future research and system enhancement are identified. **Macroeconomic Integration:** Subsequent work should embed forward-looking macroeconomic covariates into the ECL estimation framework, consistent with the full requirements of IFRS 9 [2]. GDP growth trajectories, interest rate projections, and sector-level economic indicators could be incorporated to enhance ECL prediction accuracy.

Separate PD, LGD, and EAD Modelling: Decomposing ECL estimation into its constituent parameters modelled individually and synthesised using the IFRS 9 formula would improve both interpretability and regulatory alignment. Integration of SHAP (SHapley Additive exPlanations) or LIME would further enable portfolio managers and regulators to understand the principal drivers of individual impairment predictions, enhancing model transparency [19][20].

Cross-Institutional Validation: Extending the dataset to encompass leasing records from multiple institutions would facilitate development of more broadly applicable models. **Real-Time Streaming:** Deployment within a real-time streaming pipeline using Apache Kafka would support continuous impairment monitoring as transaction events occur, enabling earlier detection of credit deterioration compared to periodic batch-based approaches.

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APPENDICES

The screenshot shows a Turnitin submission interface. At the top, there's a navigation bar with 'Class Portfolio' and a breadcrumb trail: 'CW VIEWING: HOME > FDC UNDERGRADUATE BP > UNDERGRAD THESIS CHECKING'. The main section is titled 'Submit work' with a sub-header 'Undergrad Thesis Checking ^'. It includes a 'Help' icon. The submission details are as follows:

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